Shrikaran P

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Professional Summary

Computer Science Engineering student with a strong foundation in programming, data structures and algorithms, object-oriented programming, database management, and software development. Hands-on experience in full-stack development, artificial intelligence, machine learning, cloud computing, and research-driven projects. Adapt quickly to new technologies, collaborate effectively in team environments, and apply analytical problem-solving skills to develop practical and scalable solutions.

Technical Skills

Programming Languages: Python, Java, C, C++
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP)
Web Development: HTML, CSS
Databases & Cloud: SQL, MySQL, Git, GitHub,Firebase,Google Colab
AI/ML: Machine Learning, Deep Learning, PyTorch, Scikit-learn, CNN-LSTM
Soft Skills:Team Collaboration ,Quick Learner, Time Management , Leadership, Communication

Educational Qualifications

B.Tech in Computer Science Engineering(6th semester) May 2023 - May 2027
SRM Institute of Science and Technology - Tiruchchirappalli
CGPA - 9.3/10

Achievements

- **3rd Prize Winner (Cash Award)** – Ideathon, SRM Institute of Science and Technology, for developing a Centralized Dam Management System.
- **Participant** – 24-Hour Hackathon, Dhanalakshmi Srinivasan University, where I developed a Farmer Market Price Prediction System as part of a team.
- **Foundations of AI/ML – GeeksforGeeks -** Gained hands-on experience in data preprocessing, model building, and evaluation techniques
- **Java Full Stack Development – EduSkil**l - Built web-based applications with structured data handling and backend integration

Research Internship(NIT Trichy)

- Motor Imagery EEG Classification (BCI) — Worked with BCI Competition IV Dataset 2a and BCI Competition IV Dataset 2b EEG .gdf recordings for motor imagery analysis.
- Preprocessed raw EEG signals, selected motor cortex channels, and generated STFT-based time–frequency spectrogram representations.
- Developed CNN-based models for left/right hand motor imagery classification and subject-wise performance evaluation.

Projects

Smart Interconnected Water Dam Alert System

- Coordinated multi-dam monitoring with GSM/Wi-Fi, predictive flood control, and centralized supervision.
- Implemented real-time sensor data collection and automated alert notifications to improve disaster preparedness and response.

Motor Imagery EEG Classification (BCI)

- Preprocessed .gdf EEG signals, generated STFT spectrograms, and developed a CNN-based classifier for left/right hand motor imagery.
- Improved model performance through signal preprocessing, feature extraction, and hyperparameter optimization for robust brain–computer interface applications.

Intrusion Detection System using CICIDS2017

- Developed a deep learning–based intrusion detection model to classify malicious and benign network traffic using the CICIDS2017 dataset.
- Evaluated model performance against traditional rule-based methods using accuracy, precision, recall, and F1-score metrics.

Smart Academic Timetable Generator

- Developed an automated timetable scheduling system that generates conflict-free schedules by allocating subjects, faculty, classrooms, and time slots.
- Reduced manual scheduling effort by applying constraint-based optimization to satisfy institutional and resource requirements efficiently.

Farmer Market Price Prediction System

- Built a machine learning model to predict crop market prices using historical mandi data, seasonal trends, weather conditions, and demand–supply analysis.
- Enabled farmers to make informed selling decisions through predictive analytics and data-driven market insights, improving potential profitability.

Research & Publications

- A Secure On-Premises ETL Pipeline Using OCR and Local LLMs - ICCMC 2025, Tiruppur
- Contrastive Knowledge Graph-RAG for Concept Misconceptions-RECCAP 2026, IIT Palakkad